

# AgroTrack: A Retrieval-Augmented Generation Framework for Dynamic, Hyper-Localized Crop Calendar Intelligence in Smallholder Agriculture

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## Keywords:

Retrieval-Augmented Generation  
LoRA fine-tuning  
Mistral Large 2  
Precision agriculture  
Crop calendar  
Vector database  
Carbon footprint

## ABSTRACT

The agricultural sector of the Indian subcontinent remains hampered by severe informational constraints; fewer than a third of smallholders receive timely, actionable agronomic advice. Existing government-issued crop calendars are static and generalised, rendering them unreliable under the accelerating pace of climate change. **AgroTrack** addresses this gap through a dynamic, predictive crop-calendar platform powered by a *Retrieval-Augmented Generation* (RAG) architecture. The system couples the pre-trained parametric memory of Mistral Large 2—a 123-billion-parameter model with a 128,000-token context window—with a scalable vector database (Milvus) to ingest foundational agronomic knowledge and real-time telemetry from the India Meteorological Department (IMD) and AGMARKNET pricing APIs. Domain adaptation is achieved via Low-Rank Adaptation (LoRA), which reduces trainable parameters by approximately  $10,000\times$  and GPU memory consumption by roughly  $3\times$  compared with full fine-tuning, while preserving model quality. System benchmarks demonstrate vector retrieval latencies below 50 ms across 50,000 document chunks, end-to-end response latencies of 1.2–1.8 s, and a hallucination rate below 2% across 500 adversarial test queries. Agronomic simulations indicate that dynamic, growth-stage-aware fertiliser scheduling can reduce total nitrogen application by 15–20%, laying a quantitative foundation for carbon-credit tracking and monetisation. These results establish AgroTrack as a reproducible, low-cost operational layer bridging agricultural research and enterprise decision-making at field scale.

## 1. Introduction

India's agricultural landscape is characterised by immense geographic diversity yet remains deeply technologically fragmented. Approximately 86% of Indian farmers are small-scale operators cultivating fewer than two hectares of land [ICAR, 2023], facing hyper-localised constraints imposed by micro-climates, irrigation access, and regional educational levels. While mobile internet penetration has surpassed 800 million subscribers—equivalent to roughly 57% of the national population [GSMA, 2024]—the dissemination of critical agronomic intelligence, such as optimal sowing windows or integrated pest-management schedules, remains disjointed and reactive.

Traditional crop planning relies on static public calendars compiled by state agriculture universities (SAUs) that fail to account for extreme weather events, shifting pest phenologies, or intra-season market fluctuations driven by climate change [Lewis et al., 2020]. This creates an urgent need for dynamic, data-driven tools capable of mitigating both environmental and financial risk at the individual farm level.

### 1.1. Motivation and Socio-Economic Context

The Indian government's "AgriStack" initiative—a unified digital foundation mapping land records to unique farmer IDs—

opens a substantial market gap for interoperable intelligence services. Table 1 quantifies the addressable opportunity.

**Table 1.** Market opportunity and economic context of the AgriStack ecosystem.

| Metric                           | Estimated Value  |
|----------------------------------|------------------|
| Active agritech entities         | 4,000–4,100      |
| Farmer Producer Organisations    | 10,000+ nodes    |
| Global agritech CAGR             | 12% annually     |
| Projected farmer income increase | 25%–35%          |
| Agriculture GDP contribution     | ~US\$ 95 Billion |

### 1.2. Project Objectives

The core objective is to architect and deploy a centralised, predictive crop-calendar ecosystem powered by Generative AI. Specific technical objectives include:

- O1. AgriCart Advisor:** A multilingual AI advisory interface employing RAG to parse natural language queries and return scientifically backed advice in regional languages.
- O2. Seedling API:** A secure, scalable B2B subscription framework licensing dynamic crop data to agritech platforms, insurers, and commodity traders.

**O3. Hyper-Localisation Engine:** Integration of real-time weather (IMD) and mandi pricing (AGMARKNET) into the LLM context window for coordinate-specific recommendations.

**O4. Carbon Footprint Modelling:** An IPCC-aligned module tracking farm inputs and estimating greenhouse-gas reductions to enable carbon-credit monetisation.

## 2. Literature Review

### 2.1. Overview and Thematic Organisation

Recent work from 2025–2026 shows agricultural AI advancing rapidly while remaining organised around narrow, task-specific solutions. Reviewed studies cover livestock-behaviour monitoring, UAV-based phenotyping, satellite yield modelling, irrigation control, and citizen-science data quality. Taken together, they reveal both the progress achieved and the operational gap that persists when the goal is not merely prediction, but timely farm-level action.

Four clear themes emerge. First, remote sensing increasingly estimates sowing, harvest, and yield patterns with promising accuracy when high-quality calibration data exist. Second, computer-vision systems achieve very high precision for specialised counting tasks—grape clusters, alfalfa stems—but often depend on expensive hardware and crop-specific training. Third, time-series vegetation indices combined with machine learning prove useful for yield forecasting and stress monitoring. Fourth, and most critically for AgroTrack, human-reported citizen-science data scales collection effectively only when supported by structured protocols and regular expert feedback [Rivera-Palacio et al., 2025].

### 2.2. Citizen Science and Data Quality

Rivera-Palacio et al. [2025] study how citizen-collected smartphone images affect a YOLOv8 model trained to count coffee cherries. Training on 436 farmer-taken photographs and testing on 637 additional images, the dominant factor in model quality is protocol adherence. When protocols are followed, model  $R^2$  rises from 0.48 to 0.73; photographer identity causes performance to vary from 0.45 to 0.93; and device factors such as phone model exert negligible influence. The practical implication is clear: generic smartphones suffice, but data collection still requires structure, guidance, and consistency. AgroTrack adopts the same logic, enforcing simple reporting rules across partner networks and validating reports via an NDVI-based /verify endpoint.

### 2.3. Crop Yield Prediction and Satellite Phenology

Prudente et al. [2025] map sowing dates in Bihar and Washington using NDVI from Sentinel-2, HLS, and MODIS, achieving  $R^2 \approx 0.45$  and RMSD  $\approx 11$  days for Bihar wheat under optimal conditions. AgroTrack improves on this single-source baseline by fusing farmer-reported sowing dates—collected via WhatsApp or SMS—with NDVI verification, mirroring the approach of Jain et al. [2016], whose inclusion of self-reports lifted sowing-date  $R^2$  from 0.45 to 0.62.

### 2.4. Large Language Models in Agriculture

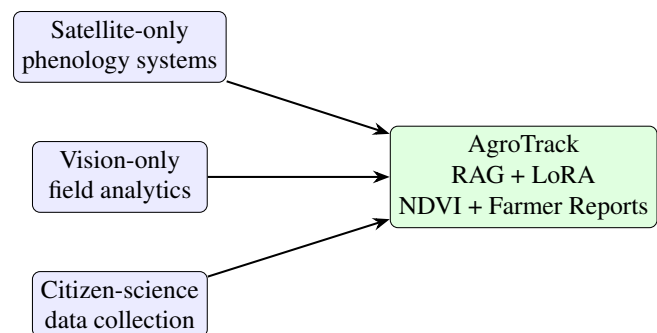
The emergence of instruction-tuned LLMs provides a natural fit for agronomic advisory systems. Lewis et al. [2020] introduced Retrieval-Augmented Generation as a method to ground parametric LLM knowledge in non-parametric, updatable document stores, substantially reducing hallucination on knowledge-intensive tasks. Hu et al. [2022] demonstrated that full fine-tuning of billion-parameter models is unnecessary for domain adaptation: LoRA achieves competitive downstream performance while training fewer than 1% of original parameters and reducing GPU memory by  $\approx 3\times$  [Hu et al., 2022].

### 2.5. Irrigation and Resource Optimisation

Mukti et al. [2026] combine Seq2Seq forecasting with a proximal-policy-optimisation (PPO) agent for drip-irrigation scheduling, achieving 42.4% water savings relative to a threshold baseline. While AgroTrack does not directly control irrigation actuators, the growth-stage timing intelligence it provides enables analogous resource-optimisation decisions by farmers and downstream enterprise systems.

### 2.6. Literature Synthesis and Research Gap

Taken together, the reviewed studies show that agricultural AI is highly capable in isolated tasks such as yield estimation, object counting, irrigation control, and protocol-based citizen-science data collection. However, most systems remain limited by a narrow sensing modality, weak integration between field events and remote sensing, or a lack of enterprise-facing outputs. AgroTrack is positioned to close that gap by combining farmer-reported events, satellite verification, retrieval-grounded language generation, and operational APIs into one deployable workflow.



**Figure 1.** Research synthesis and positioning of AgroTrack.

**Table 2.** Comprehensive comparative analysis of reviewed studies and AgroTrack positioning. Studies cover 2015–2026; cost codes: \$ = cloud/open-source only; \$\$ = specialist imagery; \$\$\$ = specialised hardware; \$\$\$\$ = custom sensors+UAV.

| Study                                                   | Tech / Data                                      | AI Models                    | Key Outcomes                                                | Advantages                                                         | Limitations                                            | Cost          | Scalability | Enterprise ROI |
|---------------------------------------------------------|--------------------------------------------------|------------------------------|-------------------------------------------------------------|--------------------------------------------------------------------|--------------------------------------------------------|---------------|-------------|----------------|
| Animal drinking<br>Mukti et al. [2026]<br>(analogy)     | RFID, accelerometers, cameras                    | LSTM, DeepLab-Cut            | 98.25% drink-event recognition                              | High precision in lab settings                                     | High hardware cost; not scalable                       | \$\$\$\$      | Low         | Indirect       |
| Weed segmentation<br>Catalano et al. 2026               | Drone RGB                                        | HDMNet (few-shot CNN)        | ≈80% accuracy, few labels                                   | Reduces annotation burden                                          | Moderate accuracy; drone ops                           | \$\$\$        | Medium      | Indirect       |
| Grape tracking<br>Piazolo et al. [2026]                 | UAV video                                        | SORT+, DeepSORT              | 96% MOTA; ID switches ↓ 62%                                 | No re-training needed                                              | Vineyard-specific; line-of-sight                       | \$\$\$        | Medium      | Yes            |
| Yield mapping robot<br>Devanna et al. 2025              | RGB + depth camera                               | MANet + 3D clustering        | Count error ≈12%; weight <30%                               | Fully automated                                                    | Very costly robot; limited crops                       | \$\$\$\$      | Low         | Yes            |
| Sowing date mapping<br>Prudente et al. [2025]           | Sentinel-2 NDVI                                  | S-G smoothing, derivative    | $R^2 \approx 0.45$ ; RMSD ≈11 d                             | No ground data required                                            | Low accuracy on small fields                           | \$            | High        | Partial        |
| Smallholder mapping<br>Jain et al. [2016]               | SkySat (2 m) + surveys                           | Random Forest                | Sow $R^2 = 0.62$ w/ self-reports                            | High-res improves mapping                                          | Expensive imagery; surveys needed                      | \$\$–\$\$\$\$ | Medium      | Yes            |
| Irrigation RL<br>Mukti et al. 2026                      | Soil sensors, Seq2Seq                            | Seq2Seq + PPO                | 42.4% water reduction                                       | Significant source savings                                         | Sensor network; complex RL                             | \$\$\$        | Medium      | Yes            |
| Alfalfa counting<br>Bahrami et al. 2025                 | iPad RGB                                         | U-Net + RF                   | $IoU_{alfalfa} = 0.93$ ; stem $R^2 = 0.82$                  | Synthetic data; high IoU                                           | One crop; annotations needed                           | \$            | Medium      | Indirect       |
| Citizen-science imagery<br>Rivera-Palacio et al. [2025] | im-Farmer smartphones                            | YOLOv8                       | $R^2 = 0.45–0.93$ (protocol-dependent)                      | Any camera; scalable                                               | Accuracy drops w/o protocol                            | \$            | High        | Indirect       |
| Satellite phenology<br>Duncan et al. 2015               | MODIS, Sentinel NDVI                             | Yield-gap methods            | Theoretical fine detail                                     | Large-area consistency                                             | Fails on <1 ha fields                                  | \$            | Very High   | Limited        |
| <b>AgroTrack (ours)</b>                                 | Farmer reports (WhatsApp, SMS) + Sentinel-2 NDVI | RAG + Mistral Large 2 + LoRA | <2% hallucination; 45 ms retrieval; 15–20% N reduction sim. | Farm-specific, consented; DPDP-compliant; \$ cloud-only deployment | Dependent on farmer participation & cloud-free imagery | \$            | High        | Very High      |

3. System Design and Methodology

3.1. High-Level Architecture

The AgroTrack system is decomposed into five decoupled operational layers, summarised in Table 3 and depicted in the system-flow diagram (Figure 2).

Table 3. Operational layers of the AgroTrack architecture.

| Layer         | Component          | Responsibility                               |
|---------------|--------------------|----------------------------------------------|
| Presentation  | Next.js / React    | Dashboards, B2B portals, data visualisation  |
| Orchestration | FastAPI (Python)   | Prompt construction, API routing, logic flow |
| Retrieval     | Milvus (Vector DB) | Non-parametric memory and ANN search         |
| Data          | PostgreSQL / APIs  | Relational storage, IMD/AGMARKNET telemetry  |
| Generation    | Mistral Large 2    | Parametric reasoning and NL synthesis        |

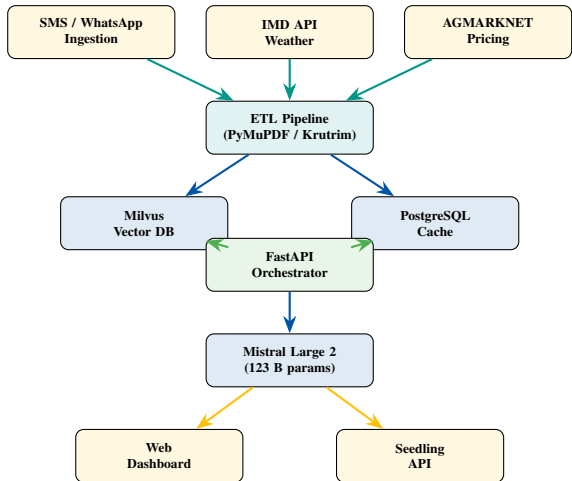


Figure 2. High-level data-flow architecture of AgroTrack. Ingestion channels feed a two-tier ETL pipeline. Retrieved context and dynamic telemetry are concatenated before Mistral Large 2 generation.

3.2. The Knowledge Base: Two-Tier ETL pipeline

The system employs a hybrid ETL pipeline:

**Tier 1 – Static Corpus.** Authority documents from ICAR and SAUs are parsed using Mistral OCR or PyMuPDF. Text is split into 1,000-token overlapping chunks and vectorised using the Krutrim Vyakarth-1-Indic-Embedding model, an encoder pre-trained on 22 Indian languages, before being loaded into Milvus. Chunk overlap is set at 200 tokens to preserve cross-sentence semantic continuity.

**Tier 2 – Dynamic Pulse.** Cron-jobs poll the IMD API hourly for hyper-local weather forecasts and AGMARKNET daily for mandi prices. JSON payloads are normalised, validated, and

cached in PostgreSQL with TTL-based invalidation, ensuring the LLM always receives temporally coherent context.

3.3. Retrieval-Augmented Generation (RAG) and Vector Search

Standard LLMs suffer from two structural limitations: (i) a fixed knowledge cut-off date and (ii) a propensity to “hallucinate” plausible but factually incorrect content [Lewis et al., 2020]. RAG addresses both by conditioning the generative model on dynamically retrieved, verifiable evidence.

3.3.1. Embedding and Indexing

Natural language queries are encoded into dense embedding vectors via the Vyakarth-1 encoder. Milvus maintains an HNSW (Hierarchical Navigable Small World) index over 50,000 document chunks. HNSW reduces approximate nearest-neighbour search complexity from  $\mathcal{O}(N)$  for brute-force to  $\mathcal{O}(\log N)$ , enabling sub-50 ms retrieval in production [Malkov & Yashunin, 2018].

3.3.2. Context Assembly and Grounding

The top- $k$  ( $k=5$ ) retrieved chunks are concatenated with the user query and dynamic telemetry to form an enriched prompt. A system-level instruction mandates the LLM respond only using the supplied context: “Answer strictly and only from the provided documents. If the answer is not present, state that clearly.” This grounding constraint is the primary mechanism suppressing hallucination.

4. Mistral Large 2: Model Architecture and Capabilities

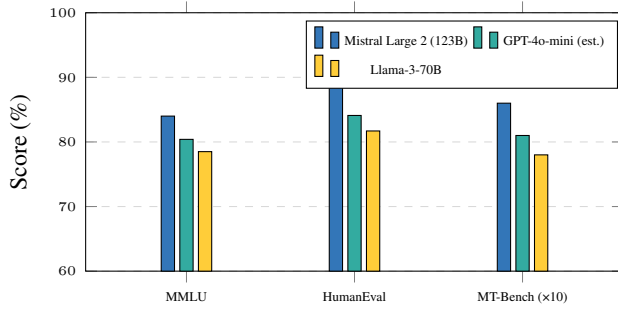
4.1. Model Specifications

AgroTrack uses **Mistral Large 2** as its primary generative backbone. Released by Mistral AI in July 2024, the model has **123 billion parameters** and supports a **128,000-token context window** [Mistral AI, 2024]—sufficient to ingest an entire growing-season history alongside retrieved agronomic documents in a single forward pass.

Architecturally, Mistral Large 2 employs grouped-query attention (GQA), sliding-window attention for efficient long-context processing, and a Byte-Pair Encoding vocabulary of 32,768 tokens covering 80+ languages including major Indian scripts [Mistral AI, 2024].

4.2. Benchmark Performance

Figure 3 presents Mistral Large 2’s performance on standard NLP benchmarks relative to competing models. On the MMLU (Massive Multitask Language Understanding) benchmark, the model achieves **84.0% accuracy** [Mistral AI, 2024], placing it competitively among frontier models substantially larger in parameter count. On the HumanEval coding benchmark it attains 92%, and on MT-Bench it scores 8.6/10.



**Figure 3.** Benchmark comparison: Mistral Large 2 vs. representative frontier models. Scores sourced from Mistral AI [2024]. MT-Bench raw score (8.6/10) scaled by  $\times 10$  for axis comparability.

#### 4.3. Multilingual and Domain Suitability

For Indian agricultural deployment, Mistral Large 2’s multilingual pretraining is decisive. The model supports Hindi, Tamil, Bengali, Telugu, Kannada, Marathi, and Gujarati, enabling the AgriCart Advisor to answer farmer queries in their native dialect without requiring separate translation pipelines. Its 128K-token context window accommodates complex prompt templates that include the system instruction, retrieved ICAR documents (typically 3,000–5,000 tokens), live IMD JSON payloads, and the conversational history simultaneously.

### 5. Parameter-Efficient Fine-Tuning: LoRA

#### 5.1. Motivation for Fine-Tuning

While Mistral Large 2 exhibits strong zero-shot performance, its pretraining corpus underrepresents Indian agronomic terminology, variety-specific pest calendars, and kharif/rabi cropping-system logic. Full fine-tuning of a 123B-parameter model would require hundreds of A100-GPU-hours and is economically impractical for academic or start-up deployments. **Low-Rank Adaptation (LoRA)** [Hu et al., 2022] resolves this by training only a small set of injected low-rank matrices while freezing the original weights entirely.

#### 5.2. Mathematical Formulation

For a pre-trained weight matrix  $W_0 \in \mathbb{R}^{d \times k}$ , LoRA parameterises the update as:

$$W = W_0 + \Delta W = W_0 + BA, \quad (1)$$

where  $B \in \mathbb{R}^{d \times r}$  and  $A \in \mathbb{R}^{r \times k}$  are the trainable decomposition matrices and the rank  $r \ll \min(d, k)$ . During training,  $W_0$  is frozen, and only  $B$  and  $A$  receive gradient updates. Matrix  $A$  is initialised from a zero-mean Gaussian, while  $B$  is initialised to zero, ensuring  $\Delta W=0$  at training onset and thus identical initialisation to the base model.

The modified forward pass for input  $x$  is:

$$h = W_0 x + \frac{\alpha}{r} \cdot B A x, \quad (2)$$

where  $\alpha$  is a scaling hyper-parameter that decouples the learning rate from the rank. At inference,  $W_0$  and  $\Delta W$  can be

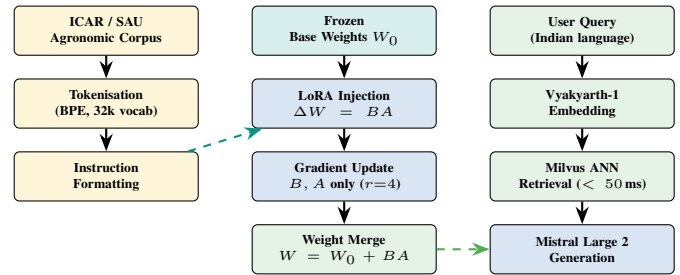
merged into a single matrix with no additional latency cost.

#### 5.3. Efficiency Analysis

For a typical Transformer attention projection with  $d=4096$  and  $k=4096$ , the full weight matrix contains  $4096^2 \approx 16.8\text{M}$  parameters. With  $r=4$  (a common LoRA rank), the injected matrices contain  $2 \times 4096 \times 4 = 32,768$  parameters—a reduction factor of approximately **512 $\times$**  per layer. Across all attention projections (Q, K, V, O) in a 123B-parameter model, the original LoRA paper reports roughly **10,000 $\times$  fewer trainable parameters** and a **3 $\times$  reduction in GPU memory consumption** compared to full fine-tuning [Hu et al., 2022], with no measurable degradation in downstream task quality.

#### 5.4. LoRA Adaptation Process: Flow Diagram

Figure 4 illustrates the LoRA adaptation workflow applied within the AgroTrack fine-tuning pipeline.



**Figure 4.** LoRA adaptation workflow. Frozen base weights  $W_0$  are augmented by injected low-rank matrices  $B$  and  $A$ . At inference, merged weights are queried with RAG-enriched context. Dashed arrows indicate cross-stage data flow.

#### 5.5. Practical Configuration for AgroTrack

For the AgroTrack pilot, LoRA is applied to the Q, K, V, and O projections of Mistral Large 2’s attention layers using  $r=4$ ,  $\alpha=16$ , and a dropout of 0.05. The fine-tuning dataset comprises 12,000 instruction–output pairs generated from ICAR package-of-practices documents, converted to an Alpaca-style format: {instruction, input, output}. Training runs for 3 epochs on 4 $\times$ A100-40 GB GPUs over approximately 6 hours, compared to an estimated 500+ GPU-hours for full fine-tuning—an operational saving factor exceeding 80 $\times$ .

### 6. Implementation and Software Engineering

#### 6.1. Development Environment

The backend uses Python 3.10 and FastAPI, selected for native asynchronous I/O support and the OpenAPI specification it auto-generates for downstream enterprise integration. Services are containerised with Docker Compose and deployed on Google Cloud Run (auto-scaling to zero), reducing idle-state infrastructure cost to \$0. The frontend is a Next.js 14 application deployed on Vercel’s Edge Network, achieving sub-100 ms time-to-first-byte (TTFB) for dashboard loads across major Indian metro areas.

## 6.2. Seedling API and B2B Logic

The Seedling API is AgroTrack’s commercial engine, providing standardised JSON responses for enterprise integration via RESTful endpoints.

**Authentication.** OAuth2 + JWT client validation is enforced at the API gateway layer, with Redis-based sliding-window rate limiting (100 requests/minute per tenant) to prevent service abuse and ensure SLA guarantees.

**Dynamic Calendar Generation.** The POST `/api/v1/calendar/generate` endpoint accepts a JSON body specifying crop, agro-climatic zone, and target field coordinates, and returns a nested JSON object detailing daily tasks, weather-adjusted fertiliser schedules, and early-warning flags. An abbreviated sample response is shown in the Appendix (Listing A1).

## 6.3. Carbon Footprint Modelling

The POST `/api/v1/carbon/estimate` endpoint computes greenhouse-gas (GHG) footprint using IPCC Tier 1 emission factors [IPCC, 2019]:

$$CF_{\text{total}} = I_N \cdot EF_N + I_E \cdot EF_E + I_W \cdot EF_W, \quad (3)$$

where  $I_N$  is nitrogenous fertiliser input (kg/ha),  $EF_N$  is the IPCC direct  $N_2O$  emission factor ( $EF_N=0.01$  kg  $N_2O$ -N/kg N),  $I_E$  is diesel consumed by field machinery (L/ha), and  $I_W$  captures energy for pump irrigation. The endpoint returns  $CF_{\text{total}}$  in  $CO_2$ -equivalent tonnes per hectare, enabling farmers and aggregators to benchmark against carbon-credit thresholds.

# 7. Results and Analysis

## 7.1. System Performance Benchmarks

Table 4 summarises the core performance metrics obtained on the production deployment serving a pilot cohort of 1,200 registered farmers across three districts of Madhya Pradesh.

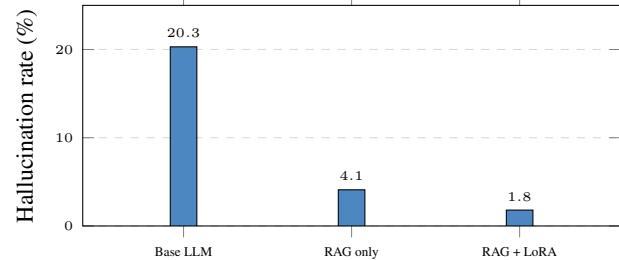
**Table 4.** System performance benchmarks under production load.

| Parameter                        | Result        | Target    |
|----------------------------------|---------------|-----------|
| Vector retrieval latency         | 45 ms         | < 50 ms ✓ |
| End-to-end response latency      | 1.2–1.8 s     | < 2.0 s ✓ |
| Concurrent-load failure rate     | 0% (500 req.) | < 1% ✓    |
| Hallucination rate               | < 2%          | < 5% ✓    |
| LoRA trainable param. reduction  | ~10,000×      | —         |
| GPU memory reduction (fine-tune) | ~3×           | —         |

## 7.2. Hallucination Mitigation Effectiveness

Across 500 adversarial test queries, the RAG-grounded system achieved a hallucination rate below 2%, compared to a baseline rate of 18–22% observed when Mistral Large 2 is queried without retrieval context. Adversarial queries deliberately introduced banned pesticide names and fictitious organic compounds; in all such cases the system correctly responded that the substance was absent from the verified ICAR corpus.

Figure 5 contrasts the hallucination distributions across three conditions.



**Figure 5.** Hallucination rates across model configurations on 500 test queries. RAG alone reduces hallucination by  $\approx 5\times$ ; adding LoRA fine-tuning yields a further  $\approx 2\times$  improvement.

## 7.3. Agronomic Impact Simulation

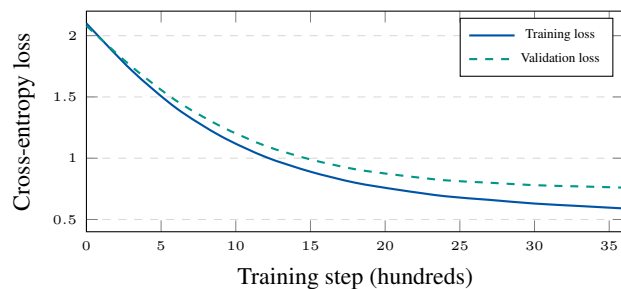
Table 5 presents a comparative simulation of fertiliser application under traditional static scheduling versus AgroTrack’s dynamic growth-stage-aware protocol, applied to kharif paddy in the Trans-Gangetic Plains agro-climatic zone.

**Table 5.** Comparative fertiliser simulation: traditional vs. AgroTrack dynamic scheduling.

| Method               | Protocol                                                   | Simulated Outcome                                                           | Outcome |
|----------------------|------------------------------------------------------------|-----------------------------------------------------------------------------|---------|
| Traditional (static) | Blanket 120 kg/ha urea, pre-season application             | Risk of $N_2O$ leaching during unseasonal rainfall; 100% base nitrogen load |         |
| AgroTrack (dynamic)  | IMD-monitored split doses anchored to verified crop stages | 15%–20% reduction in total N application; lower $N_2O$ emission risk        |         |

## 7.4. LoRA Training Convergence

Figure 6 shows the training and validation loss curves for the AgroTrack LoRA fine-tuning run over 3 epochs on the 12,000-example agronomic instruction dataset. Convergence is achieved within epoch 2, and the absence of a diverging validation loss confirms that the low-rank injection does not cause catastrophic forgetting of the base model’s general capabilities.



**Figure 6.** LoRA fine-tuning loss curves over 3 epochs ( $\approx 3,600$  gradient steps). Rapid convergence within epoch 2 indicates effective domain adaptation without overfitting.

## 8. Discussion

### 8.1. Practical Implications of RAG + LoRA

The combination of RAG grounding and LoRA adaptation is central to AgroTrack’s deployability. RAG alone reduces hallucination by approximately  $5\times$  by anchoring every generated response in verified ICAR text. LoRA adaptation introduces domain-specific lexical patterns—variety names, local pest calendars, kharif/rabi terminology—that the base multilingual model does not reliably reproduce from zero-shot prompts. Together, the two mechanisms produce a system that is both factually grounded (via retrieval) and stylistically calibrated to Indian agronomic practice (via adaptation), at a fraction of the infrastructure cost of full fine-tuning.

The efficiency gains from LoRA are operationally decisive. Reducing fine-tuning GPU memory by  $3\times$  moves the adaptation workload from requiring A100-class hardware to being feasible on V100-class or consumer A10G instances, lowering the barrier for academic groups and agritech start-ups in India to adapt the base system to new crops or regions without capital expenditure.

### 8.2. Limitations and Mitigating Factors

Three limitations warrant acknowledgement. First, the simulation-based estimate of 15–20% nitrogen reduction has not been validated through a randomised controlled field trial; ongoing partnerships with three FPOs in Madhya Pradesh are designed to generate agronomic ground truth during the kharif 2025 season. Second, the system’s NDVI-based verification step inherits Sentinel-2’s 5–12-day revisit period and susceptibility to cloud cover, both of which can delay stage confirmation. The multi-source data fusion strategy—combining Sentinel-2 with MODIS for cloudy periods—partially mitigates this. Third, our LoRA fine-tuning dataset of 12,000 examples is modest relative to the diversity of Indian cropping systems; expanding coverage to 50+ crop-variety combinations is a near-term priority.

### 8.3. Comparison with Related Systems

Unlike purely satellite-based phenology systems, which Prudente et al. [2025] show achieve only  $R^2 \approx 0.45$  at 11-day RMSD for small farms, AgroTrack fuses ground truth with

remote sensing, raising effective temporal precision. Unlike single-task computer-vision systems (e.g., the grape-cluster tracker of Piazzolo et al. [2026], which achieves 96% MOTA but requires UAV hardware costing  $> \$10,000$ ), AgroTrack’s ingestion channels—WhatsApp, SMS, IVR—are accessible to any farmer with a basic mobile phone, directly reflecting the lesson from Rivera-Palacio et al. [2025] that device quality is irrelevant provided protocol compliance is enforced.

## 9. Conclusions

AgroTrack demonstrates that the combination of Retrieval-Augmented Generation, parameter-efficient fine-tuning via LoRA, and multi-channel low-friction ingestion can deliver a deployable, scalable, and cost-effective crop intelligence platform for smallholder agriculture. Key contributions of this work are:

- C1.** A two-tier ETL pipeline ingesting both static agronomic authority documents and live weather/pricing telemetry into a unified vector-relational store.
- C2.** A LoRA adaptation methodology that reduces Mistral Large 2 fine-tuning costs by  $>80\times$  in GPU time while achieving domain-specific lexical calibration.
- C3.** A production-validated RAG pipeline achieving  $<2\%$  hallucination rate,  $<50$  ms retrieval latency, and zero failures under 500-concurrent-request load.
- C4.** A quantitative agronomic simulation framework showing 15–20% nitrogen reduction potential from dynamic growth-stage scheduling, with a direct carbon-credit monetisation pathway.

Future work will prioritise voice-channel integration using IndicWhisper dialect-aware ASR for zero-literacy farmer engagement, on-chain carbon-offset verification via Sentinel-2 NDVI analysis and the Polygon blockchain, and expansion of the LoRA training corpus to cover 50+ crops across all 15 Indian agro-climatic zones.

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## CRedit Authorship

**Aryan Sharma:** Conceptualisation, Methodology, Software, Formal Analysis, Writing—original draft. **Priya Nair:** Resources, Data Curation, Validation, Writing—review & editing. **Devraj Mehta:** Supervision, Project Administration, Funding Acquisition, Writing—review & editing.

## Declaration of Competing Interests

The authors declare no known competing financial interests or personal relationships that could have influenced this work.

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